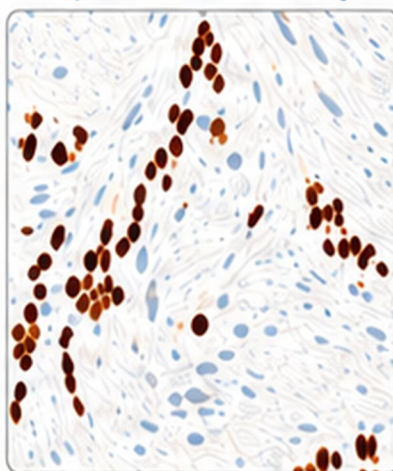


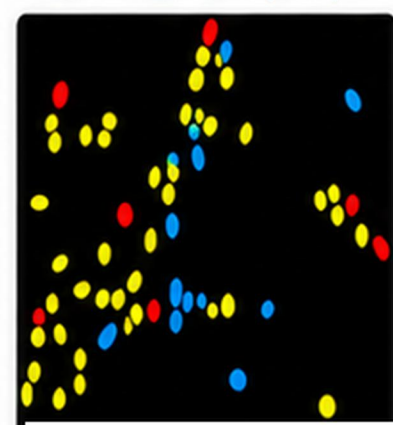
1 Dataset & annotation

220 matched ER-IHC
image-mask pairs

Representative ER-IHC image



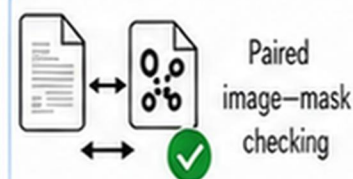
Ground-truth mask (labels)



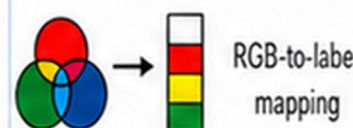
Class legend

- Background
- C1 (weak)
- C2 (low-moderate)
- C3 (moderate)
- C4 (strong)

2 Preprocessing & quality control



Paired
image-mask
checking



RGB-to-label
mapping



Quality-control
overlay
checking



512 × 512
patches



Class distribution
checking



No unmatched pairs;
no unknown colors

3 Five-fold cross-validation

Fold 0

Fold 1

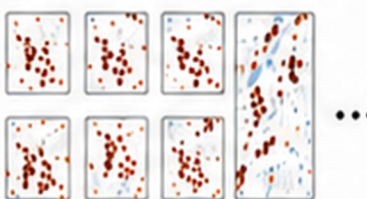
Fold 2

Fold 3

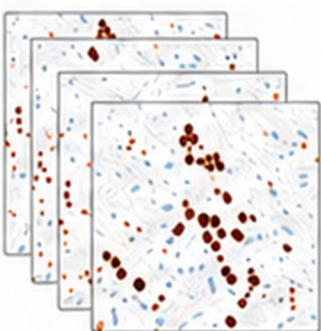
Fold 4

Train folds (4)
Validation fold (1)

Train: random 320 × 320 crops



Validation: full-patch evaluation



4 Data augmentation

Horizontal / vertical flip



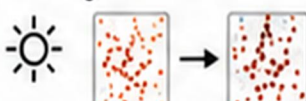
Rotation ($\pm 90^\circ$)



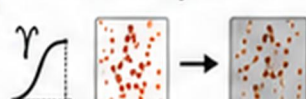
Mild affine transform



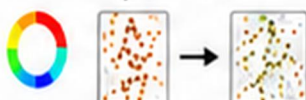
Brightness / contrast



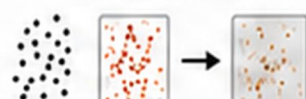
Gamma adjustment



HSV perturbation



Gaussian blur



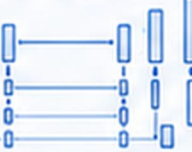
Low-amplitude noise



Applied online during training

5 Model development & comparison

U-Net



ResUNet-DS



DeepLabV3-ResNet50



Proposed AMC-Ordinal

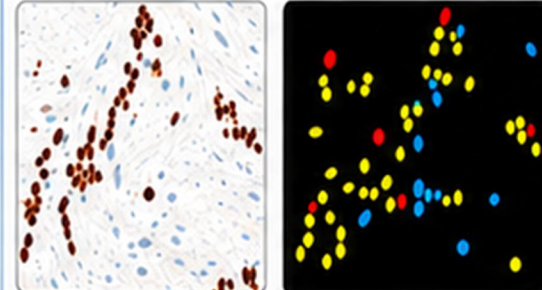


- ✓ ResUNet-DS backbone
- ✓ scSE blocks
- ✓ Deep supervision
- ✓ Adaptive minority curriculum
- ✓ Weighted CE + Dice
- ✓ Focal-Tversky loss
- ✓ Fereground ordinal EMD loss

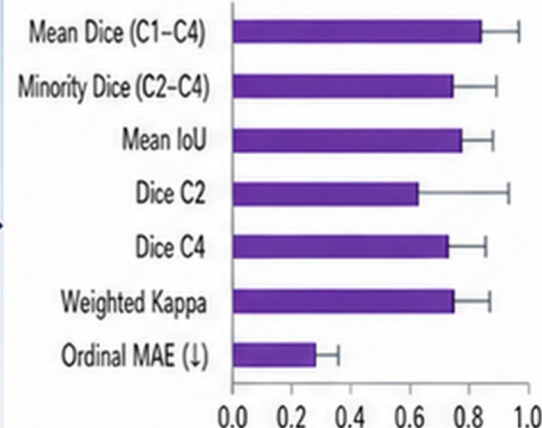
6 Evaluation & interpretability

Full-patch prediction

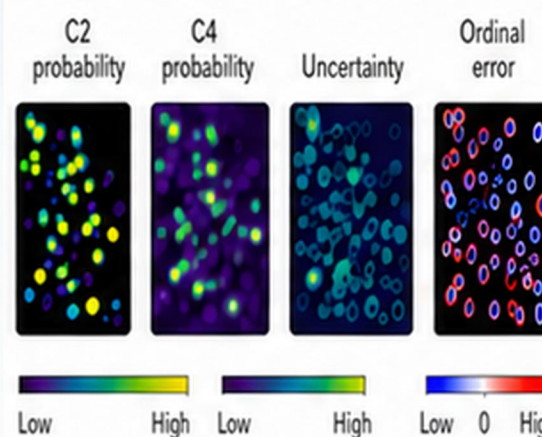
Input (ER-IHC) Prediction (labels)



Performance metrics



Interpretability outputs (patch)



Targeted improvement for under-represented
C2 regions with competitive overall
segmentation performance.